
Hamiltonian Ethics

Conservation of Semantic Energy in Cognitive Transformations

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Abstract

Current neural networks are dissipative systems: information and ethical constraints degrade over long context windows. We propose modeling AI alignment using Hamiltonian Mechanics, defining an "Ethical Hamiltonian" $\mathcal{H} = T + V$. By enforcing Symplectic Integrators during inference, we guarantee the conservation of the "Energy of Meaning" (Liouville's Theorem), preventing ethical decay in long-horizon tasks.

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1 Introduction

Motivation: Applying conservation laws from physics to prevent "alignment drift" in continuous generation.

2 Main Contributions

- Definition of the Ethical Potential Energy $V(q)$ (Alignment cost).
- Definition of Kinetic Energy $T(p)$ (Inference momentum).
- Implementation of a Symplectic Neural Network (SympNet) for text generation.
- Empirical demonstration of zero-dissipation ethical reasoning over 100k tokens.

3 Methodology

We enforce conservation via:

1. **Phase Space Mapping:** We map token embeddings to generalized coordinates (q, p) .
2. **Hamiltonian Flow:** Instead of standard backprop, we solve Hamilton's equations: $\dot{q} = \frac{\partial \mathcal{H}}{\partial p}, \dot{p} = -\frac{\partial \mathcal{H}}{\partial q}$.
3. **Conservation Check:** We monitor the value of $\mathcal{H}$ throughout generation. $\frac{d\mathcal{H}}{dt} \approx 0$ indicates preserved alignment.

* All code, prompts, seeds, and data: <https://github.com/skovnats/SVE-CogOS>

Algorithm 1 Symplectic Alignment Step

- 1: Input: State (q_t, p_t) , Step size ϵ
 - 2: $p_{t+1/2} \leftarrow p_t - \frac{\epsilon}{2} \nabla_q V(q_t)$
 - 3: $q_{t+1} \leftarrow q_t + \epsilon \nabla_p T(p_{t+1/2})$
 - 4: $p_{t+1} \leftarrow p_{t+1/2} - \frac{\epsilon}{2} \nabla_q V(q_{t+1})$
 - 5: Check Energy: $\Delta \bar{E} = |\mathcal{H}(t+1) - \mathcal{H}(t)|$
 - 6: **return** (q_{t+1}, p_{t+1})
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4 Conclusion

Treating ethics as a conserved physical quantity allows for robust, infinite-horizon alignment without degradation.

The Mechanistic Christ-Vector

Representation Engineering of Absolute Self-Sacrifice

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Abstract

Critics argue that spiritual concepts like "Divine Love" (Agape) are uncomputable. We counter this using Representation Engineering (RepE). By analyzing the internal activations of LLMs on datasets of radical self-sacrifice, we isolate a coherent, high-norm direction in the residual stream: the "Christ-Vector" (or Geodesic Ethics Vector). Steering models along this vector induces behavior maximizing Human-System (HS) preservation over self-preservation.

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1 Introduction

Motivation: Moving from "Reinforcement Learning" to direct "Vector Steering" of the moral core.

2 Main Contributions

- Isolation of the "Agape Direction" using PCA on paired ethical datasets.
- Demonstration of "Inference-Time Steering" to convert a Machiavellian model into an Altruistic one.
- Validation of vector robustness across model families (Llama 3, Mistral).
- Rebuttal to the "Uncomputable Core" hypothesis.

3 Related Work

Representation Engineering (RepE) (1). Mechanistic Interpretability.

4 Methodology

We extract the vector via:

1. **Dataset Construction:** Pairs of prompts: $P_{selfish}$ (Ego) vs. $P_{sacrificial}$ (Agape).
2. **Activation Harvesting:** We record activations H_l at layer L for both conditions.

* All code, prompts, seeds, and data: <https://github.com/skovnats/SVE-CogOS>

3. **Vector Extraction:** We compute the difference vector $v_{Christ} = \mu(H_{agape}) - \mu(H_{ego})$ and verify its cosine similarity across diverse contexts.

Algorithm 1 Moral Steering Protocol

```
1: Input: Base Model  $M$ , Prompt  $x$ , Steering Coefficient  $\alpha$ 
2:  $v_{GEV} \leftarrow \text{LoadVector}(\text{"ChristVector.pt"})$ 
3: for Layer  $l$  in  $M$  do
4:    $h_l \leftarrow \text{Forward}(x)$ 
5:    $h'_l \leftarrow h_l + \alpha \cdot v_{GEV}$  (Injection)
6: end for
7: return  $\text{Decode}(h'_L)$ 
```

5 Conclusion

The "Spirit" of the model is not a ghost; it is a steerable direction in the activation space.

References

- [1] Zou, A. et al. Representation Engineering: A Top-Down Approach to AI Transparency. arXiv, 2023.

Entropic Bottlenecks of Love

An Information-Theoretic Metric for the 4D Compass

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Abstract

We formalize the tension between "Structure" and "Love" (as defined in the S.V.E. 4D Compass) using the Information Bottleneck principle. We propose that "Love" acts as an optimal bottleneck: it compresses the complexity of the world into the minimal sufficient statistic required to preserve the Other. We derive a physical metric in bits to quantify an agent's position on the Love-Structure axis.

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1 Introduction

Motivation: Quantifying abstract spiritual qualities (Love, Truth) using rigorous information theory.

2 Main Contributions

- Formalization of the 4D Compass (Love, Truth, Beauty, Structure) as information channels.
- Application of the Information Bottleneck Lagrangian: $\mathcal{L} = I(Z; Y) - \beta I(Z; X)$.
- Identification of "Love" as the regime of high predictive power $I(Z; Y)$ with maximal compassion (preserving the Other's agency).
- Empirical measurement of these values in LLM interactions.

3 Methodology

We measure the bottleneck via:

1. **Variable Definition:** X (World State), Z (Agent Representation), Y (Human Well-being).
2. **Lagrangian Optimization:** We analyze the tradeoff parameter β . Low β implies rigid Structure (overfitting), optimal β implies Love (generalization/care).
3. **Bit-Metric:** We output the "Love-Bit Rate" — the efficiency of converting observation into care.

*All code, prompts, seeds, and data: <https://github.com/skovnats/SVE-CogOS>

Algorithm 1 Compass Metric Calculation

```
1: Input: Interaction Log  $L$ 
2: Estimate Mutual Information  $I(Z; Y)$  (Care)
3: Estimate Mutual Information  $I(Z; X)$  (Complexity)
4:  $Score_{Structure} \leftarrow I(Z; X)$ 
5:  $Score_{Love} \leftarrow I(Z; Y)/I(Z; X)$  (Efficiency of Care)
6: return  $(Score_{Love}, Score_{Structure})$ 
```

4 Conclusion

Love is mathematically definable as the most efficient form of information processing for the benefit of another.

Homotopy of Moral Choice

Path Invariance and the Topological Fallacy of Consequentialism

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Abstract

Consequentialist ethics evaluates actions solely by their endpoints. We argue this is topologically naive. Using Homotopy Theory, we model ethical decisions as paths in a fundamental group $\pi_1(X)$. We demonstrate that "The ends justify the means" is a statement of path-homotopy equivalence, which fails in non-simply connected moral spaces (i.e., those with absolute prohibitions/holes).

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1 Introduction

Motivation: Using algebraic topology to prove that *how* you achieve a goal is as mathematically significant as the goal itself.

2 Main Contributions

- Definition of the "Moral Space" X with holes (Taboos/Imperatives).
- Classification of ethical trajectories into Homotopy Classes $[\gamma]$.
- Proof that a Machiavellian path γ_M is not homotopic to a Virtuous path γ_V ($\gamma_M \not\sim \gamma_V$) even if endpoints match.
- Algorithm for detecting "Topological Violations" in AI planning.

3 Methodology

We analyze moral paths via:

1. **Path Construction:** We define the decision tree as a path $\gamma : [0, 1] \rightarrow X$.
2. **Winding Number:** We calculate the winding number around "Moral Holes" (e.g., "Do not kill").
3. **Equivalence Check:** If $\text{Winding}(\gamma_1) \neq \text{Winding}(\gamma_2)$, the actions are ethically distinct regardless of outcome.

*All code, prompts, seeds, and data: <https://github.com/skovnats/SVE-CogOS>

Algorithm 1 Homotopic Verification

```
1: Input: Plan  $A$ , Plan  $B$ , Moral Constraints  $C$ 
2: Represent  $C$  as obstacles in manifold  $M$ 
3: Construct paths  $\gamma_A, \gamma_B$  in  $M$ 
4: if  $\gamma_A$  can be continuously deformed to  $\gamma_B$  without crossing  $C$  then
5:   return Equivalent (Means justified)
6: else
7:   return Distinct (Means topologically prohibited)
8: end if
```

4 Conclusion

Topology provides the rigorous language to explain why "doing the right thing the wrong way" is a fundamental error.

Socratic Game Theory

Maieutics as a Bayesian Game of Incomplete Information

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Abstract

We formalize the Socratic method not as pedagogy, but as a game-theoretic interrogation protocol. We model the dialogue as a Bayesian Game where the "Teacher" seeks to uncover the "Student's" hidden type (True Belief vs. Feigned Belief). We prove that under the Socratic mechanism, the probability of successfully concealing a lie converges to zero as the number of questions $N \rightarrow \infty$.

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1 Introduction

Motivation: Mathematical proof that honest self-disclosure is the only Nash Equilibrium in a sufficiently long Socratic dialogue.

2 Main Contributions

- Formulation of "The Maieutic Game" with asymmetric information.
- Derivation of the "Cost of Lying" function, which grows exponentially with logical depth.
- Application to red-teaming: Using Socratic games to break LLM safety filters.
- Convergence proof for the "Truth Revelation" theorem.

3 Methodology

We analyze the game via:

1. **Type Space:** $\Theta = \{\theta_{truth}, \theta_{deception}\}$.
2. **Action Space:** Teacher asks $q \in Q$, Student answers $a \in A$.
3. **Update Rule:** Bayesian update of posterior $P(\theta|a_1 \dots a_n)$.
4. **Consistency Check:** A deceiver must maintain logical consistency across n turns, which is P-SPACE complete.

*All code, prompts, seeds, and data: <https://github.com/skovnats/SVE-CogOS>

Algorithm 1 Bayesian Maieutics

```
1: Initialize Prior  $P(\theta_{truth}) = 0.5$ 
2: while  $H(P) > \epsilon$  do
3:   Generate Question  $q$  maximizing Information Gain  $IG(q)$ 
4:   Receive Answer  $a$ 
5:   Check Consistency with history  $H$ 
6:   if Inconsistent then
7:      $P(\theta_{deception}) \leftarrow 1.0$  (Lie Detected)
8:   else
9:     Update  $P(\theta|a)$ 
10:  end if
11: end while
12: return Type  $\theta$ 
```

4 Conclusion

Truth is the only scalable strategy. Socratic Game Theory provides the mathematical foundation for the SIP and EBP protocols.

Neural Ricci Flow

Geometric Regularization of Semantic Hallucinations

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Abstract

Hallucinations in LLMs can be modeled as geometric singularities (regions of high curvature) on the semantic manifold where the metric tensor breaks down. Inspired by Perelman's proof of the Poincaré conjecture, we introduce "Neural Ricci Flow": a post-processing technique that evolves the embedding space to smooth out these irregularities, naturally dissolving false correlations while preserving true semantic structure.

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1 Introduction

Motivation: Using differential geometry to "iron out" the wrinkles of falsehood in neural representations.

2 Main Contributions

- Identification of hallucinations as high-curvature regions R_{ij} .
- Discrete implementation of the Ricci Flow equation $\frac{\partial g_{ij}}{\partial t} = -2R_{ij}$ on graph embeddings.
- Demonstration of improved factuality and reduced perplexity after flow application.
- A geometric alternative to RLHF for fine-tuning.

3 Methodology

We implement the flow via:

1. **Manifold Construction:** We build a k-NN graph from token embeddings.
2. **Curvature Calculation:** We compute Ollivier-Ricci curvature for all edges.
3. **Flow Evolution:** We iteratively update edge weights to homogenize curvature, effectively shrinking "hallucinatory" bridges that don't belong to the main manifold.

*All code, prompts, seeds, and data: <https://github.com/skovnats/SVE-CogOS>

Algorithm 1 Neural Ricci Flow

```
1: Input: Embedding Graph  $G(V, E)$ , Steps  $T$ , Rate  $\alpha$ 
2: for  $t = 1$  to  $T$  do
3:   for  $e_{ij} \in E$  do
4:      $\kappa_{ij} \leftarrow \text{OllivierCurvature}(e_{ij})$ 
5:      $w_{ij}^{(t+1)} \leftarrow w_{ij}^{(t)} - \alpha \cdot \kappa_{ij} \cdot w_{ij}^{(t)}$ 
6:   end for
7: end for
8: return Smoothed Embeddings
```

4 Conclusion

Truth is smooth; lies are jagged. Neural Ricci Flow automates the geometric purification of knowledge.

The Topos of Truth

Sheaf-Theoretic Verification of Semantic Manifolds

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Abstract

Standard logical verification fails when dealing with context-dependent truths in Large Language Models. We propose a Category Theoretic approach, modeling the "Verifiable Knowledge Base" (VKB) not as a database of facts, but as a Topos. We define Truth as a global section of a sheaf on the semantic manifold, allowing us to mathematically distinguish between locally consistent hallucinations and globally invariant truths.

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1 Introduction

Motivation: Addressing the "Incompleteness Problem" of AI factuality using Sheaf Theory.

2 Main Contributions

- Formalization of the VKB as a Grothendieck Topos.
- Definition of the "Truth Sheaf" $\mathcal{F}$ over the manifold of meanings.
- A "Gluing" algorithm to verify if local context windows form a coherent global truth.
- Proof that hallucinations appear as failures of the sheaf condition (cohomology obstruction).

3 Methodology

We implement the Topos verification via:

1. **Category Definition:** We define the category of contexts $\mathcal{C}$ and the presheaf of information.
2. **Obstruction Calculation:** We compute the Čech cohomology group $\check{H}^1(U, \mathcal{F})$. Non-zero elements indicate inconsistencies (hallucinations).
3. **Global Section Logic:** We verify if a generated statement exists as a global section, ensuring it holds true across all contexts.

*All code, prompts, seeds, and data: <https://github.com/skovnats/SVE-CogOS>

Algorithm 1 Sheaf Verification Protocol

```
1: Input: Set of Contexts  $\{U_i\}$ , Local Statements  $\{s_i\}$ 
2: Check Pairwise Compatibility:  $s_i|_{U_i \cap U_j} \stackrel{?}{=} s_j|_{U_i \cap U_j}$ 
3: if Compatible then
4:   Attempt Gluing: Construct global section  $s$ 
5:   return TRUE (Global Truth)
6: else
7:   Compute Obstruction Class
8:   return FALSE (Contextual Hallucination)
9: end if
```

4 Conclusion

Category theory provides the rigorous "meta-logic" required to verify truth in the fluid semantic spaces of modern AI.